Procurement, Rev B build with the bought track belt

Date 16 Oct 2026. Prices S$ ex GST as seen in listings; confirm at checkout. Rev B = Nhien's pack (`RB4107 Rev B`), whose BOM covers everything in section 2. This is the version with Nhien's belt (BOM row B10) instead of my printed track; everything else is identical to the printed-track list.

1. Track side (bought belt, Rev B B10)

Part	Qty	Make / buy	Where	Price	Arrives
Track belt HTD 550-5M-15, 110 teeth, 15 wide (e.g. SKF PHG 550-5M-15)	4	**Buy**	RS SG, stock no. 058-6535 or MISUMI SG HTD 5M belts	see listing (about S$25-35 each)	RS SG stock: 1-3 days
Cleats, rubber 10 x 7.9 x 15, bonded on the belt back every 5th tooth (25 mm), 22 per belt	88	**Buy + bond**: belt shop	Local belt / conveyor shop (Fenner, Megadyne or Gates distributor), with the belts	quote	allow 1-2 weeks
Deck change for the belt: notch P12 13 x 135 over each front lane (x -25 to 110), or cut the cleats to 5 high	1	**Make**: with the deck laser cut, or file / mill after	Professional Workshop	-	with the deck
Timing pulley HTD 5M 20T, 15 mm belt, flanged, **plain (pilot) bore**	8	**Buy**	MISUMI SG, Timing Pulleys HTD 5M	~S$20 each	ships 1 day from SG stock; allow 2-5 days
Pulley bore modification: 4 x D-bore Ø8 (M1), 4 x Ø16 H7 (M2)	8	**Make**: Professional Workshop request, Rev B sheet 21	E3-01-06	free	workshop queue, allow 1-2 weeks

2. Nhien's (frame, drive, hinge, electronics), from the Rev B BOM

Part	Qty	Make / buy	Where
Side plates P1 (front) and P2 (rear), 5 mm Al, laser + tap	2 + 2	Make	DXF in the pack; Professional Workshop or a laser shop on the vendor list
Slide rails P3-P6, UHMW strip 189 x 20 x 5	8	Buy + drill	MISUMI SG UPA-189-20-5
Rail brackets RS PRO 427-991 (angle 15 x 14.6 x 10)	32	Buy	RS SG stock no. 427-991
Hinge shafts P20 A 82.3 + B 77.3 (or one 212), Ø8 D-shaft flat 7	2	Buy stock, cut	RS SG linear shafts or MISUMI SG
Bearing 688-2RS 8 x 16 x 5	14	Buy	RS SG ball bearings
Shoulder screw ISO 7379 Ø8 x 25 M6 + washers, shim, spacer, nyloc	4 sets	Buy	RS SG shoulder screws
2020 profile x 110 (P7) x2, 3030 x 160 (P8) x2, corner posts P15 x4	1 m + 0.5 m	Buy, cut	MISUMI SG / local
Corner brackets P18 (25x25x4 angle) x4, P19 (20x20x3) x4, T-nuts, E-clips, shims	-	Buy / make	Rev B BOM sections 3-4
Deck P12, tray P16, cover P22, motors, drivers,	-	Theirs	Rev B BOM

ESP32, battery			

3. What changes against the printed-track version

No TPU filament, no fit-test slice, no printing. Instead: 4 belts from RS, a belt-shop job for the 88 cleats (the long pole, 1-2 weeks), and the deck notch so the cleats clear the deck. The wheels, bore modification, plates, shafts, bearings and shoulder screws are the same in both versions, so they can be ordered before the track choice is final.

4. Order of events

1. Order the 8 plain-bore pulleys and the 4 belts today (critical path, then the workshop bore mod and the belt shop); send the belts with the cleat drawing (TS-01B) to the belt shop.
2. Nhien: laser plates and the notched deck, order rails, bearings, shoulder screws, profiles.
3. Workshop: pulley bores, plate tapping, shaft cutting (one request form).
4. Assemble per Rev B sheet 3; fit the cleated belts on the real pulleys; tension at the slots.